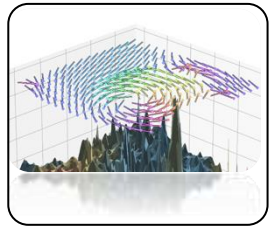
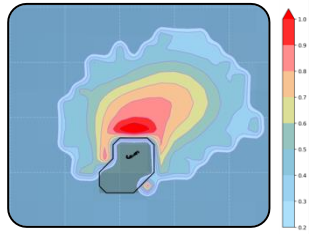
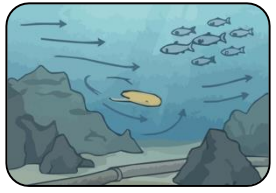
Data-Grounded Ocean Environment



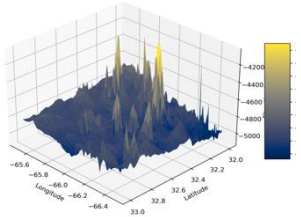
Currents & Tides



Pollution & Elements



Marine Organisms



Bathymetry & Terrain



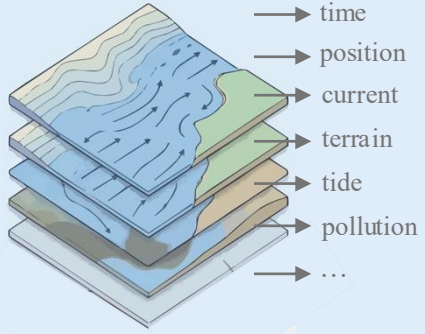
Environment Query Interface & Tasks Design

- Navigation / Tracking
- Sensing / Inspection
- Coordination / Pollution

- *same tasks*
- *same tiered difficulty*
- *same grounding logic*
- *comparable across backends*

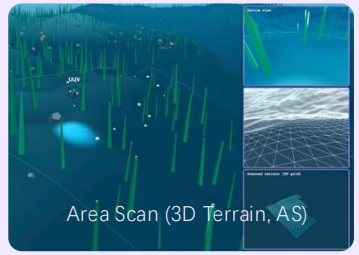
Tier 1 | Tier 2 | Tier 3

Ocean Signal=$\boldsymbol{Q}$ (*position, time, depth, context*)



Observations O_t

Canonical Task Ladder + Agent Interaction



Area Scan (3D Terrain, AS)



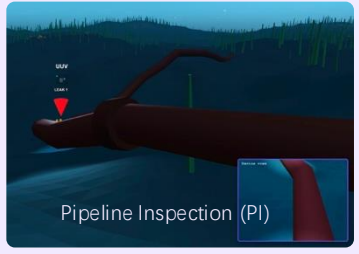
Underwater Pollution Lift (UPL)



Fish Protection / Herding (FH)



Route Following (RF)



Pipeline Inspection (PI)



Surface Pollution Cleanup (SPC)

Policy / Controller

- Heuristic
- Learned, LLM high-level planner + low-level controller

Optional Memory / Planning State

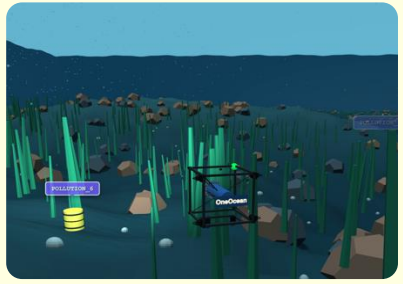
Action a_t

- bounded control command

Same task contract across both backends

Agent Interaction

Core Benchmark Backend



3–6 DoF dynamics | 10 Tasks,
Single-agent / Multi-agent (2-10)

External Scene Backend



Standardized Outputs

- Standardized Metrics
- Success Rate / Tracking Error
- Episode Time / Energy Usage
- Collisions /Violations
- Coordination Cost